

Shailesh Chaudhari

Software Engineer - Backend and Distributed Systems

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SUMMARY

Software Engineer with 2.5+ years building backend and distributed systems: multi-tenant test-execution infrastructure, idempotent webhook pipelines, and real-time platforms. Works in TypeScript and Node.js with PostgreSQL, Redis, and Docker. Strong focus on correctness: concurrency control, deterministic testing, and verification that asserts real outcomes.

EXPERIENCE

Software Engineer, ContextQA

Jul 2025 - Present

- Build the backend of the core QA-automation platform: a test-execution engine orchestrating concurrent runs across Playwright, WebDriverIO, and LambdaTest; VNC streaming for live browser-session debugging; and the REST APIs driving run history, assertions, and artifacts.
- Shipped 2 Chrome extensions in the first months: Vibe Testing (AI-assisted UI testing with real-time log streaming over WebSockets) and AxeTos (accessibility auditing against WCAG A/AA/AAA with automated fix suggestions). Both shipped and still in use.

Software Developer, eSparkBiz Technologies

Aug 2024 - Jul 2025

- Delivered 3 simultaneous client products full-stack (EdTech scheduling, sign-language e-learning, corporate LMS), all on schedule with zero production incidents.
- Integrated a third-party scheduling API with a Python/CPLEX optimization service; synchronized large datasets and built a real-time calendar UI backed by AWS S3 file workflows.

Software Developer Intern, eSparkBiz Technologies

Jan 2024 - Aug 2024

- Built REST APIs and Next.js/NestJS front-ends with authentication and role-based access control (RBAC); integrated Stripe payments and Supabase authentication.

PROJECTS

KhataGO - WhatsApp-first bookkeeping for Indian MSMEs khatago.vercel.app

- Idempotency-guarded reconciliation pipeline: Meta webhook to database-enforced deduplication to LLM function-calling to ledger writes to Tally XML export. Exactly-once effects over an at-least-once delivery channel, enforced by a unique constraint rather than a distributed lock; 8 simultaneous deliveries of the same message produce exactly one ledger write.
- Containerized (Docker) polling worker over a PostgreSQL-backed queue with an atomic claim, exponential backoff with jitter, and a dead-letter queue (DLQ) with audited replay.

BALLAST - deterministic simulation of a multi-tenant session control plane github.com/Shailesh93602/ballast

- Per-tenant concurrency caps, quota windows, and leases over an unreliable substrate; every run is a pure function of one integer seed and replays byte-identically, verified across 1,000 seeds.
- Verified 4 independent ways: 8 invariants, a reference oracle, mutation testing (100 of 116 mutants killed), and a self-verifying shrinker. Found 6 real bugs, half of them in the checker rather than the implementation.
- Measured 3 concurrency-control strategies for a flash sale against a real PostgreSQL under 200 concurrent buyers: read-then-write oversold 200 of 5 units; conditional update and optimistic versioning each sold exactly 5.

EduScale - real-time coding battles across multiple Node.js instances eduscale.vercel.app | github.com/Shailesh93602/DevScale

- Cross-instance coordination across 2+ Node.js server instances for real-time battles: Socket.io Redis adapter for pub/sub, Redlock distributed mutex for battle start, per-key circuit breakers around LLM calls, and Prometheus metrics.

SKILLS

- **Languages:** TypeScript, JavaScript, SQL, C++
- **Backend:** Node.js, Express.js, NestJS, Next.js, REST APIs, Socket.io, Prisma ORM
- **Databases:** PostgreSQL, Redis, MongoDB (relational and non-relational)
- **Infrastructure:** Docker, Docker Compose, GitHub Actions (CI/CD), Vercel, Supabase, AWS S3
- **Reliability patterns:** idempotency, concurrency control, distributed locks, message queues and dead-letter queues, circuit breakers, mutation and property-based testing

EDUCATION

- B.E. in Information Technology, Gujarat, India - 2020 to 2024 - CGPA 7.99/10

ACHIEVEMENTS

- Institute Rank 1 on GeeksforGeeks
- 700+ problems solved on CodeChef; 5-star C++ rating on HackerRank
- Finalist, New India Vibrant Hackathon 2023